



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-7: 2 Classes

*Please Listen Carefully
&*

Keep your phone(s) on silent mode

Chemical Kinetics
Bioenergetics
Biocatalysis

Association and dissociation constants



The general reaction form is: $\mathbf{aA + bB \rightarrow cC + dD}$

Reactions are categorized as zero-order, first-order, second-order, or mixed-order (higher-order) reactions.

- **Zero-Order Reactions** (where order = 0)

Have a constant rate. The rate of a zero-order reaction is constant and independent of the concentration of reactants. The rate law is: $\text{rate} = k$, with k having the units of M/sec.

- **First-Order Reactions** (where order = 1)

Have rate proportional to the concentration of one of the reactants. Eg: radioactive decay, the spontaneous process through which an unstable atomic nucleus breaks into smaller, more stable fragments.

The rate law is: $\text{rate} = k[A]$ (or B instead of A), with k having the units of sec^{-1}

- **Second-Order Reactions** (where order = 2)

Have a rate proportional to the concentration of the square of a single reactant or the product of the concentration of two reactants.

The formula is: $\text{rate} = k[A]^2$, with the units of the rate constant $\text{M}^{-1}\text{sec}^{-1}$

The binding constant, or association constant, is a special case of the equilibrium constant K , and is the inverse of the dissociation constant. It is associated with the binding and unbinding reaction of receptor (R) and ligand (L) molecules, which is formalized as: $R + L \rightleftharpoons RL$ (R: Receptor and L: Ligand)

The reaction is characterized by the on-rate constant k_{on} and the off-rate constant k_{off} , which have units of $\text{M}^{-1} \text{s}^{-1}$ and s^{-1} , respectively. In equilibrium, the forward binding transition $R + L \rightarrow RL$ should be balanced by the backward unbinding transition $RL \rightarrow R + L$. That is,

$$k_{\text{on}} [R] [L] = k_{\text{off}} [RL]$$

Association and dissociation constants



The binding constant K_a is defined $K_a = \frac{k_{\text{on}}}{k_{\text{off}}} = \frac{[\text{RL}]}{[\text{R}] [\text{L}]}$

An often-considered quantity is the dissociation constant $K_d \equiv 1/K_a$, which has the unit of concentration, despite the fact that strictly speaking, all association constants are unitless values.

- k_{off} is the rate constant of dissociation of the drug from the receptor; k_{on} is the rate constant of association of the drug to the receptor.
- **Dissociation constant** (K_d) is the rate constant of dissociation at equilibrium, defined as the ratio $k_{\text{off}} / k_{\text{on}}$
- **Association constant** (k_a or K_a) is the opposite of K_d
 - When K_a is high, K_d is low, and the drug has a high affinity for the receptor (fewer molecules are required to bind 50% of the receptors)
- **Affinity** in chemistry is the tendency of dissimilar chemical species to form chemical compounds. **The major factors which affect affinity and dissociation constant in clinical pharmacology are temperature and the presence of a catalyst.** The major factors which affect affinity and dissociation constant are temperature and the presence of a catalyst

Association and dissociation constants



In general, the rate of any reaction is determined by the Arrhenius equation:

$$k = Ae^{\frac{-E_a}{RT}}$$

k is the rate constant

E_a is the activation energy for the reaction

R is the universal gas constant;
 T is the absolute temperature in Kelvin

A is the pre-exponential factor constant, which is different for every chemical reaction, and which equates to the frequency of molecular collisions in the correct orientation

$$R = 8.3145 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$$

So, it is composed of several immutable constants, eg. the gas constant and temperature. Within the confines of the fragile reaction vessel which is the human body, temperature is fairly constant and probably close to 37°C. The only things which will vary for any given chemical reaction will be E_a , which can be reduced in the presence of a catalyst.

Association and dissociation constants

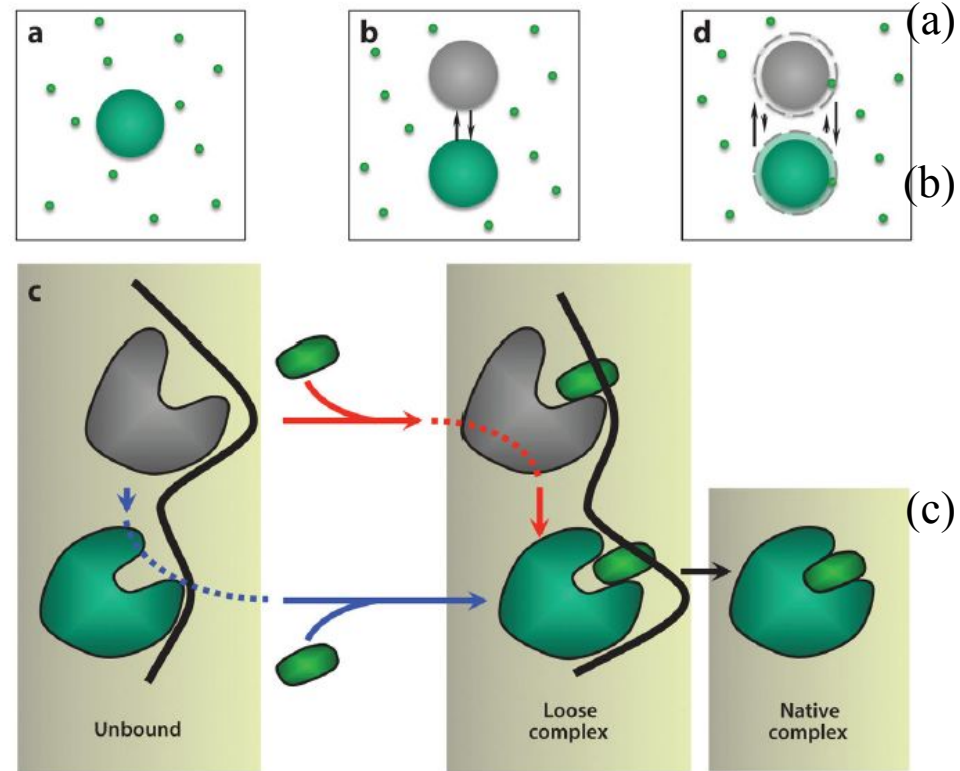


The function of a catalyst is to lower the activation energy required by a reaction. Therefore, the lowered activation energy (accounted by the catalyst) can be substituted into the Arrhenius equation in order to obtain the rate constant for the catalysed reaction.

The exponential part of the Arrhenius equation ($-E_a/RT$) accounts for an exponential increase in the value of the rate constant for any decrease in the activation energy. Since the rate of a chemical reaction is directly proportional to the rate constant of that reaction, the decrease in activation energy results in an exponential increase in the reaction rate.

It is important to note that the rates of uncatalysed reactions are more severely affected by temperature than the rates of catalysed reactions. This is because the activation energy is in the numerator of the exponential term $-E_a/RT$ and the absolute temperature is in the denominator. Since the activation energy of the catalysed reaction is relatively low, the effect of temperature on the rate constant is more visible in the corresponding

Protein-ligand binding



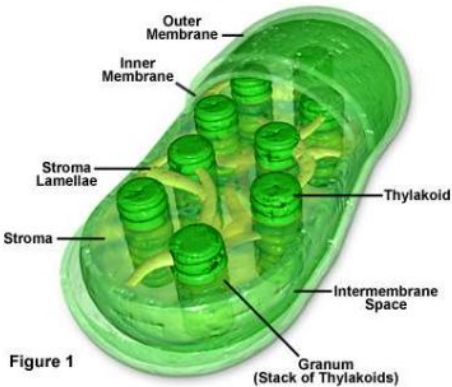
- (a) A spherical protein surrounded by point-like ligands. The binding reaction occurs as soon as a ligand comes into contact with the protein; this scenario is modeled mathematically by an absorbing boundary condition on the protein surface.
- (b) The protein undergoing indifferent switch between two conformations. The interconversion rates between the two conformations are the same regardless of where the ligands are. In the inactive conformation (gray sphere), the binding reaction cannot proceed; this situation is modeled by a reflecting boundary condition on the protein surface. The active conformation (cyan sphere) allows the binding reaction to proceed, as modeled by the absorbing boundary condition.
- (c) Alternative pathways via which the inactive unbound protein reaches the native complex. In the conformational-selection pathway (blue arrows), the protein first makes the inactive-to-active transition and then loosely binds a ligand. In the induced-fit pathway, the order of these two steps is reversed (red arrows). The loosely bound complex converts to the native complex upon further tightening of the protein–ligand interactions (black arrow).
- (d) The protein undergoing induced switch. The interconversion rates depend on whether there are free ligands.

The dissociation constant is commonly used to describe the affinity between a ligand (such as a drug) and a protein i.e., how tightly a ligand binds to a particular protein. Ligand-protein affinities are influenced by **non-covalent intermolecular interactions between the two molecules such as hydrogen bonding, electrostatic interactions, hydrophobic and van der Waals forces**. Affinities can also be affected by high concentrations of other macromolecules, which causes macromolecular crowding. The formation of a ligand-protein complex LP can be described by a two-state process where [P], [L], [LP] represent molar concentrations of the protein

$$K_d = \frac{[L][P]}{[LP]} \quad \text{complex respectively} \quad L + P \rightleftharpoons LP$$

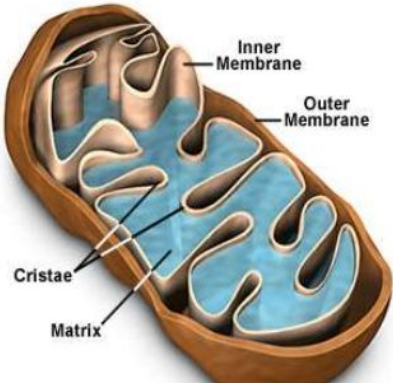
Bioenergetics

- Identify and describe the cell structures involved in processing energy.
- Describe the fundamental roles of plastids (e.g., chloroplasts) and mitochondria in energy transformations.
- Plastids** – A group of membrane-bound organelles commonly found in photosynthetic organisms; in control of synthesis and storage of food.
Ex. Chloroplast, Amyloplast (store starch).
- Plant Cell Chloroplast Structure**



Chloroplast – An organelle found in plant cells and the cells of other eukaryotic photosynthetic organisms where photosynthesis occurs.

Mitochondria Structural Features



Mitochondria – A membrane-bound organelle found in most eukaryotic cells including plants; site of cellular respiration.

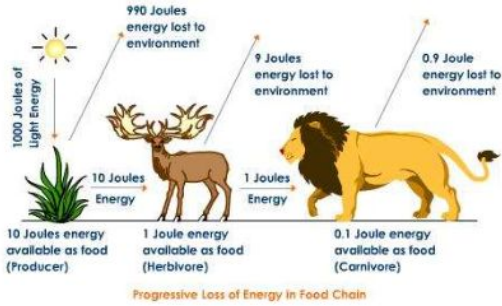
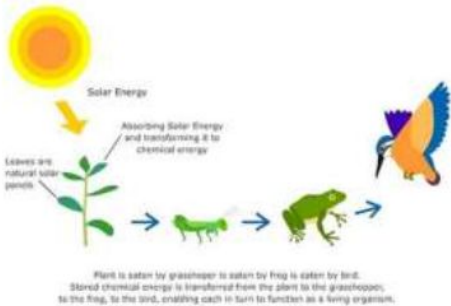
Identify and describe how organisms obtain and transform energy for their life processes.

Bacteria/Archea	Protists	Fungi	Plants	Animals
Glycolysis or chemosynthesis by absorbing chemicals through the plasma membrane and processing them in the cytoplasm	Glycolysis by absorbing chemicals from the environment, photosynthesis with chloroplasts or cellular respiration with mitochondria	Absorb chemicals from a host, cellular respiration with mitochondria	Photosynthesis with chloroplasts to store light energy from the sun in carbohydrates, then cellular respiration with mitochondria.	Consumption of carbohydrates and lipids from the environment, then digestion, absorption, circulation and respiration with mitochondria.

- Compare the basic transformation of energy during photosynthesis and cellular respiration.

Bioenergetics – The study of energy flow (energy transformations) into and within living systems.

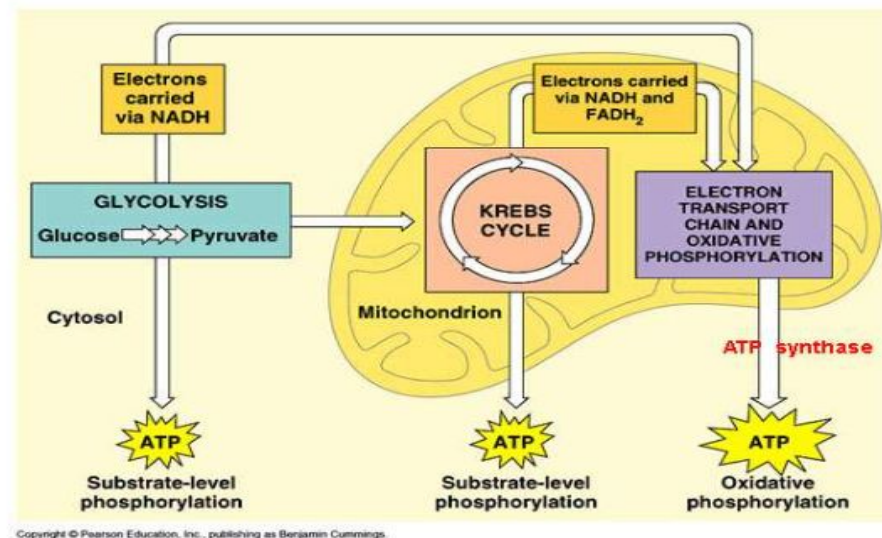
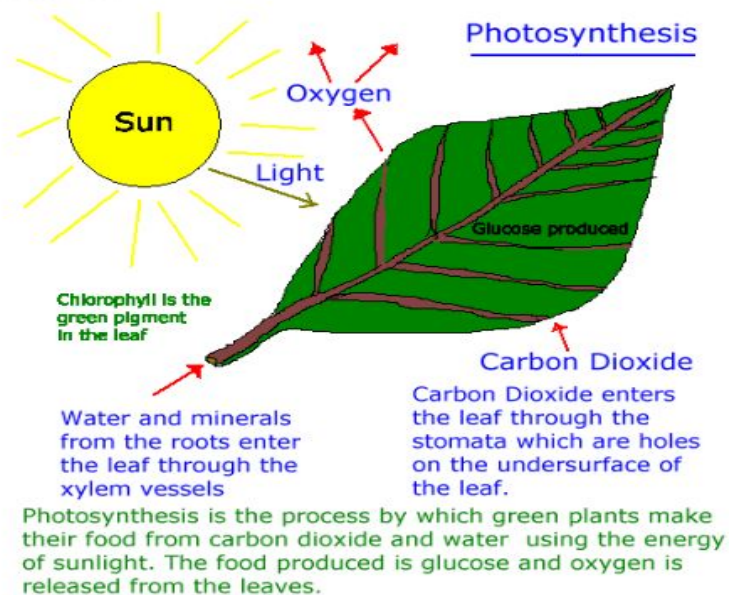
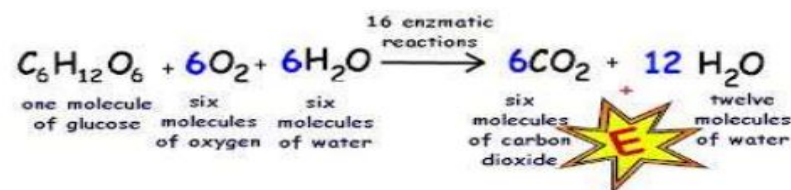
Energy Transformation – A process in which energy changes from one form to another form while some of the energy is lost to the environment.



- Bioenergetics** refers to the study of energy flow and transformation in living organisms. This includes the processes of energy generation, storage, and utilization that are essential for life. Living organisms require energy to carry out various processes such as growth, reproduction, and movement.
- The **primary source of energy** for most living organisms is sunlight, which is converted into chemical energy through the process of photosynthesis in plants and some bacteria. This chemical energy is then transferred to other organisms through food chains and food webs.
- It is the quantitative study of energy transductions in living cells.
- Types of energy: Kinetic energy (energy in motion) and Potential energy (stored

Detailed examples of Bioenergetics

Biochemical Conversion – The changing of organic matter into other chemical forms such as fuels.

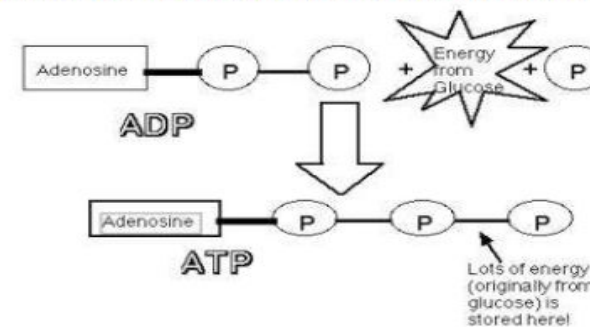
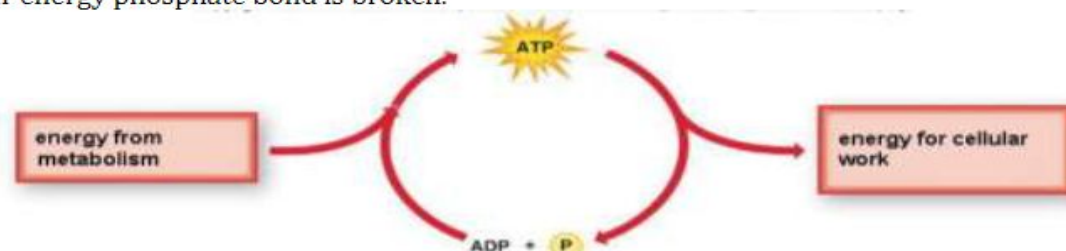


- Photosynthesis
- Cellular Respiration

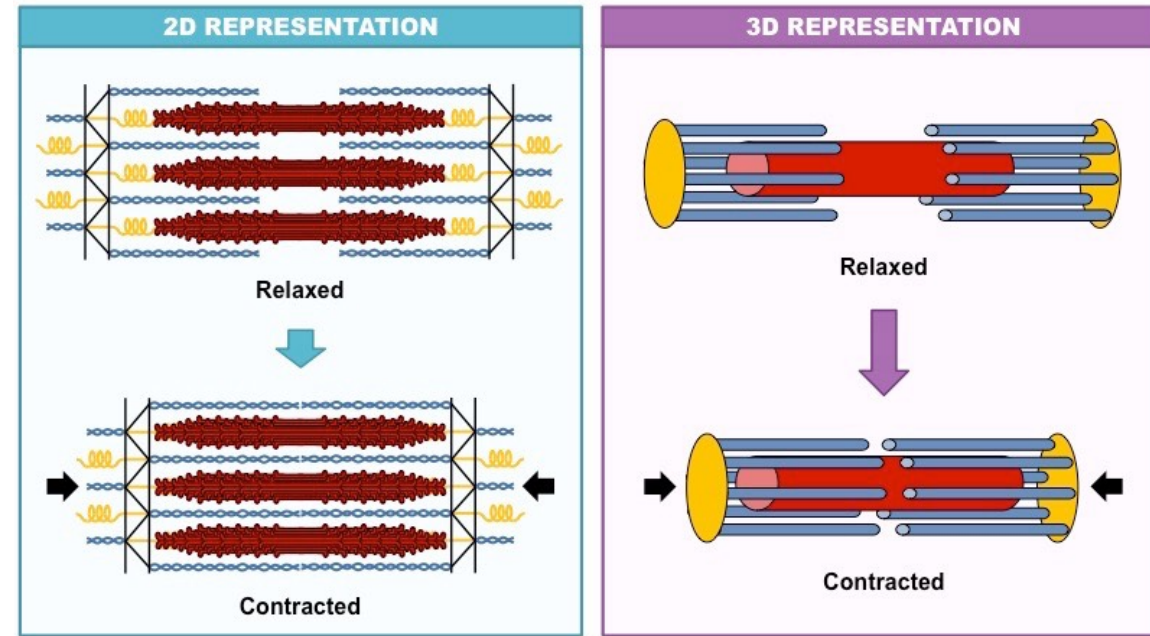
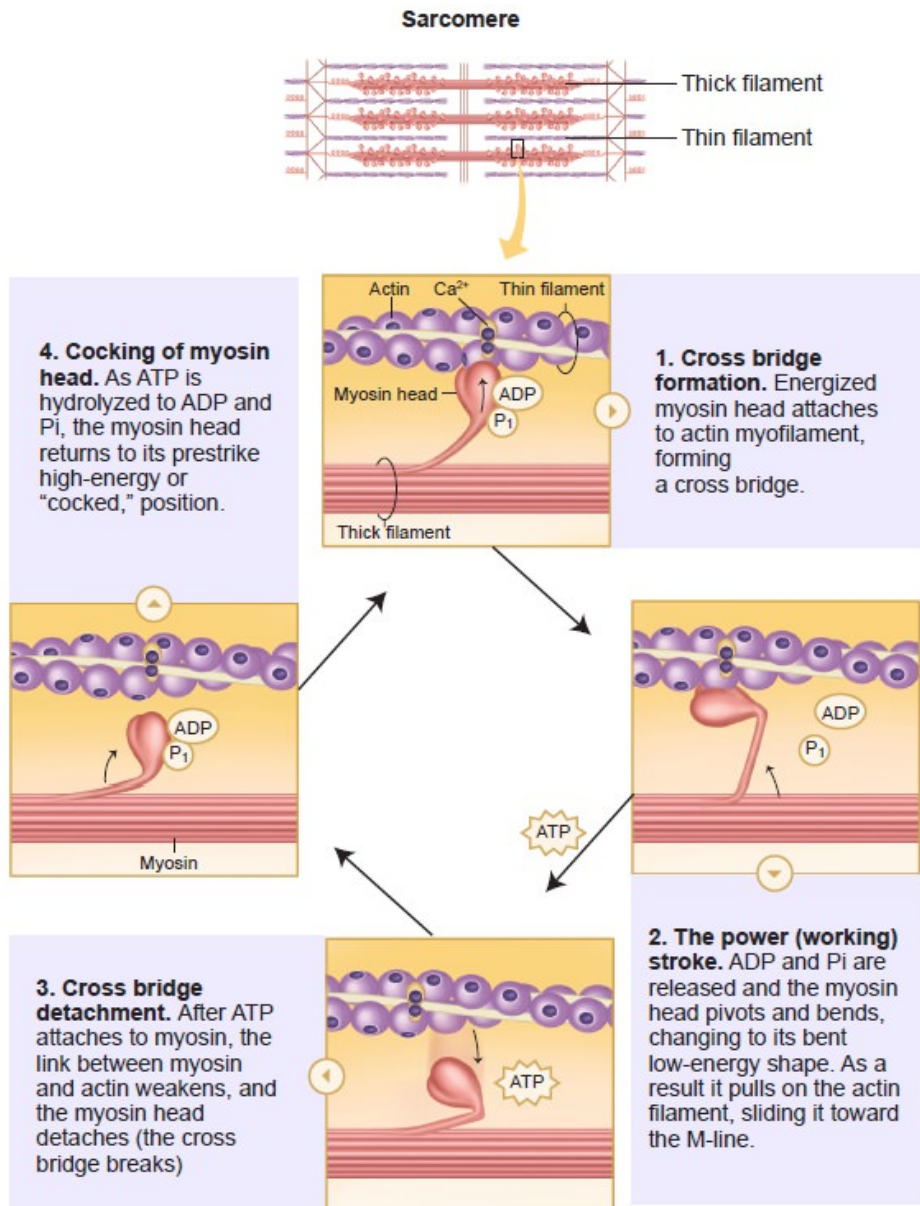
Photosynthesis – A process in which solar radiation is chemically captured by chlorophyll molecules and stored in the bonds of carbohydrate molecule.

- Describe the role of ATP in biochemical reactions.

Adenosine Triphosphate (ATP) – A molecule that provides energy for cellular reactions and processes. Energy is released when a high-energy phosphate bond is broken.



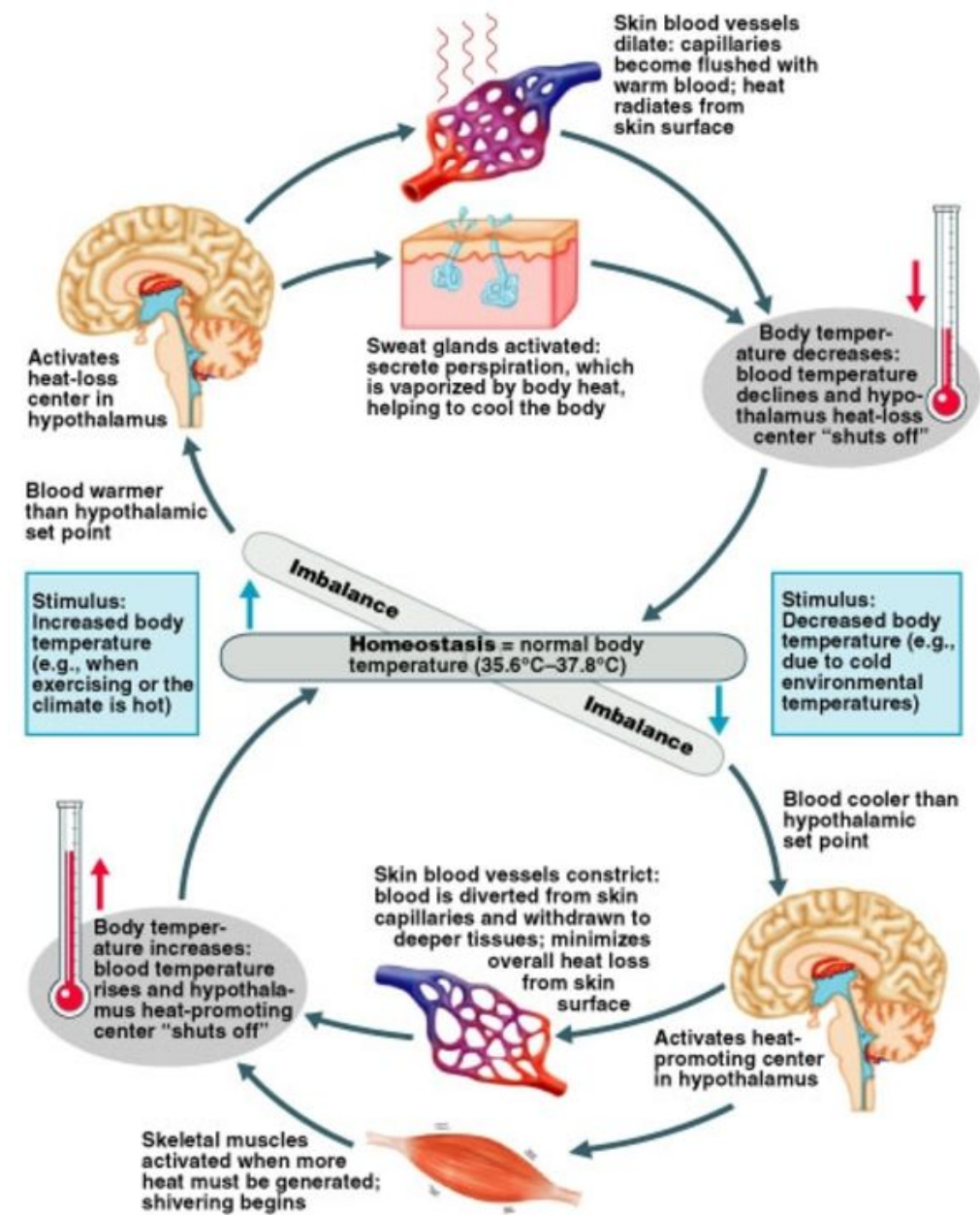
Detailed examples of Bioenergetics – Muscle contraction



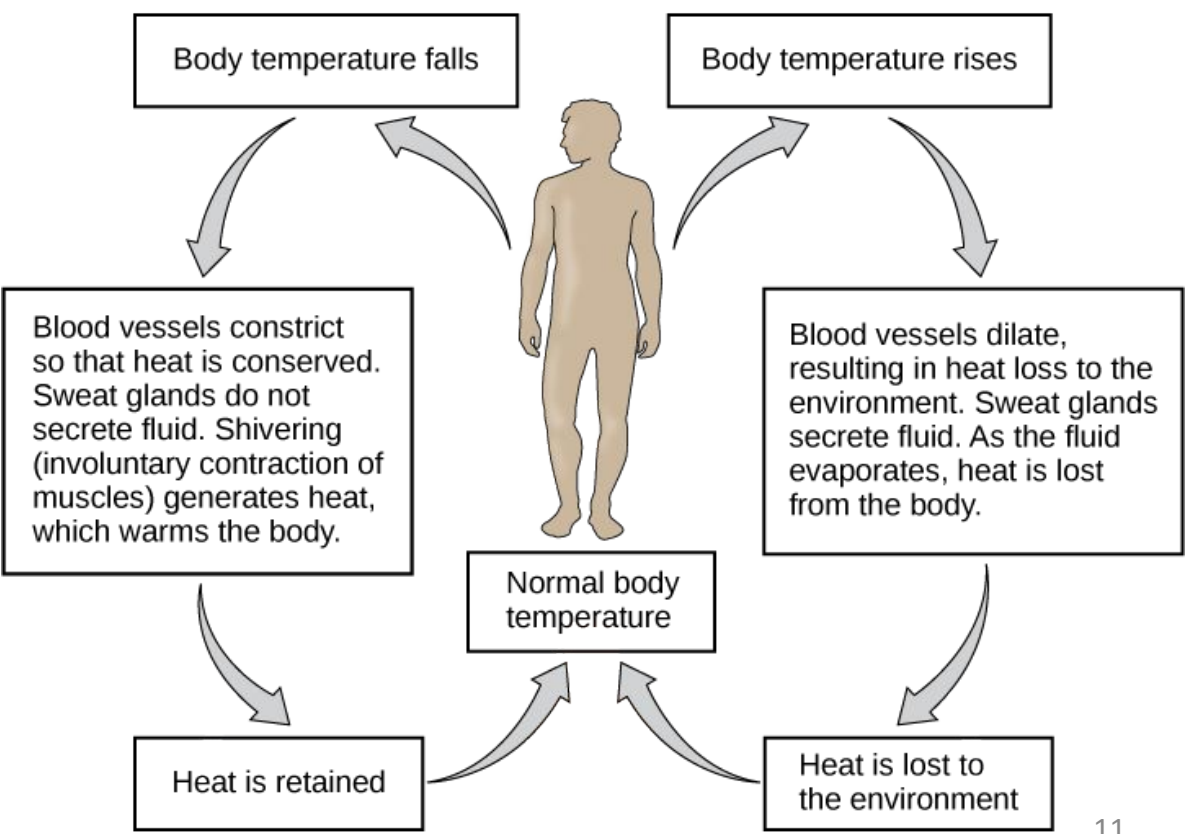
- Muscle cells require large amounts of energy to contract and produce movement. This energy is provided by the breakdown of ATP, which is stored in the muscle cells.
- The process of muscle contraction involves the conversion of chemical energy stored in ATP into mechanical energy, which is used to move the

Figure 9.4 Cross-bridge cycle of muscle contraction

Detailed examples of Bioenergetics – Thermoregulation



Maintaining a constant body temperature requires energy. This process involves the generation and dissipation of heat through various mechanisms such as sweating, shivering, and changes in blood flow. In cold environments, organisms may use metabolic processes to generate heat and maintain their body temperature.

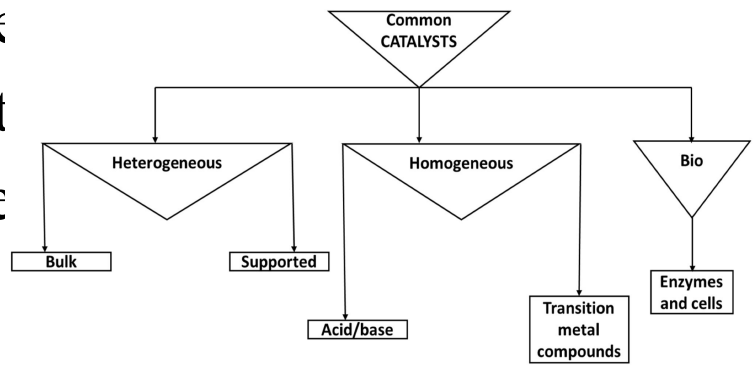


Biocatalysis

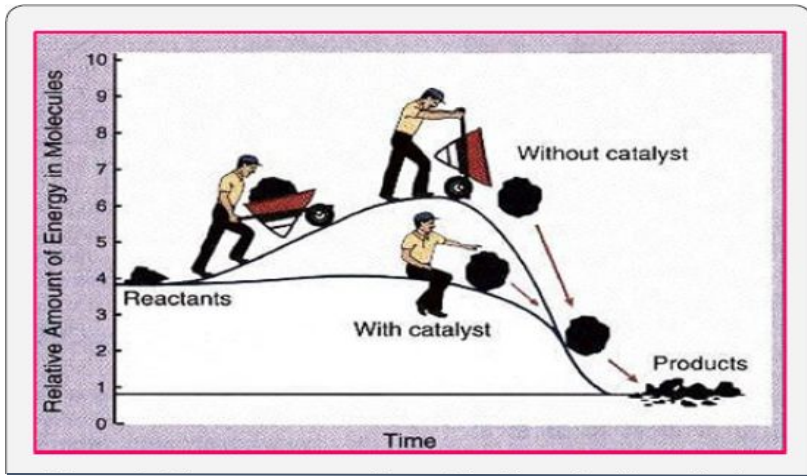
Biocatalysis- Altering the rate of biological reactions!

Catalyst-

A chemical/substance/entity that alters the rate of a reaction and doesn't get changed chemically and quantitatively after the reaction.



Processes 2022, 10(4), 683



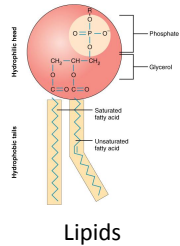
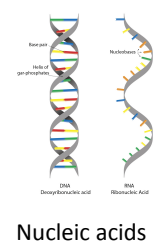
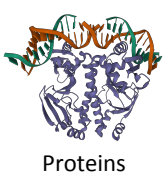
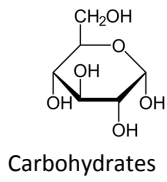
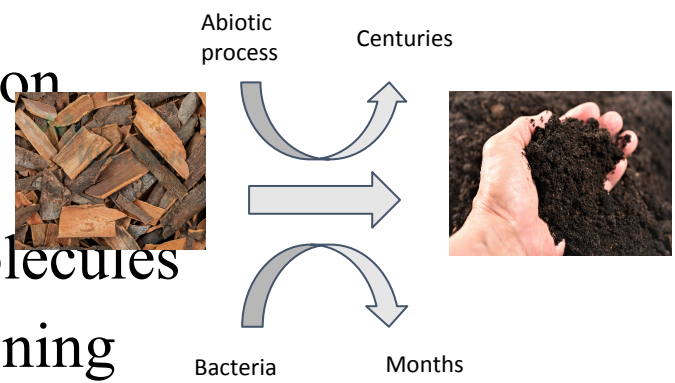
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Biocatalysts-

Catalysts present in the biological system.

The cells themselves are considered to act as biocatalysts. For e.g. decomposition of organic matter.

However, in the end, it's all about biomolecules as they work for the survival and functioning of cells.



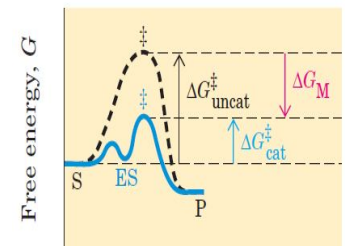
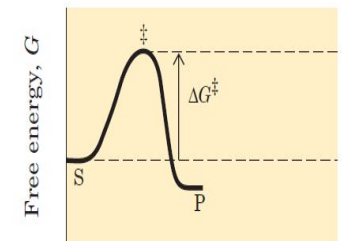
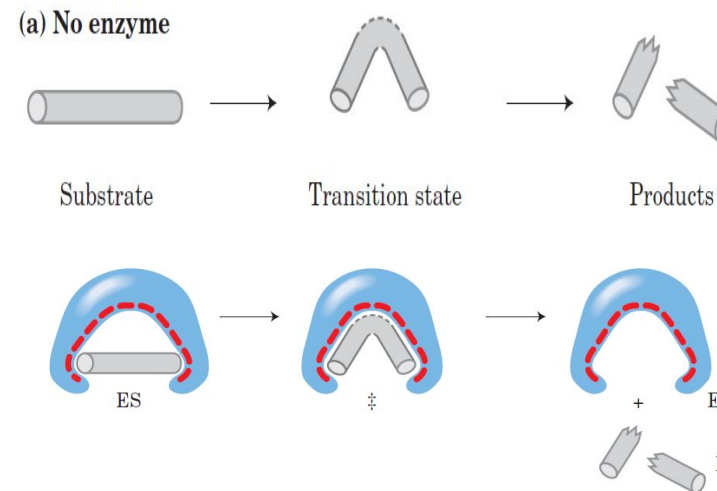
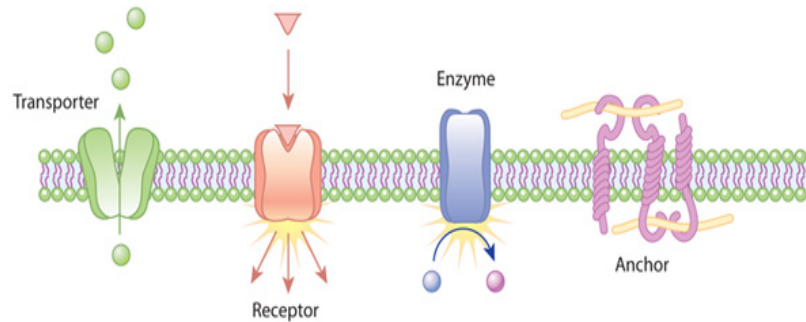
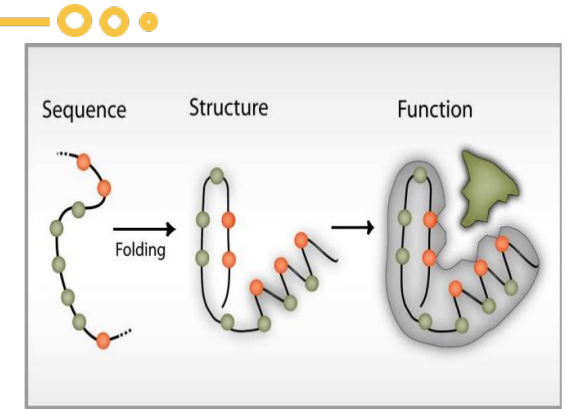
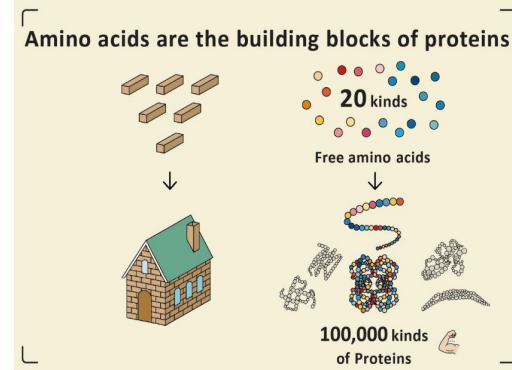
Biocatalysis

In biological systems, **proteins** (made up of amino acids) are the major functional molecules which perform diverse functions ranging from signalling to catalysis.

The **diversity in structure** of proteins allows them to acquire different locations and **functions** in an organism. The class of **proteins that take part in catalysis of reactions are known as Enzymes**. The reactants whose catalysis are done are called **substrates**. Few of the properties of enzymes include-

- Complex protein molecules with high M.W.
- Very specific in their action.
- Maximally active at pH 6-8.
- Are highly regulated by several factors.
- Highly efficient and reusable.

A collection of amino acids at a particular site makes an “Active site” of enzyme where the catalysis reaction takes place. The **weak binding interactions between the active site and the substrate** provide a substantial driving force for enzymatic catalysis.



Nitrogen Fixation

According to NASA, the total composition of air in the earth's atmosphere consists of 78% nitrogen gas, 21% of oxygen, 0.93% argon, 0.04% carbon dioxide, and small amounts of other gases.

Many cellular components are also made up of nitrogen. Despite all of it, the concentration of a usable form of nitrogen is very less. Nitrogen fixation is the essential biological process and the initial stage of the nitrogen cycle. In this process, nitrogen in the atmosphere is converted into ammonia (another form of nitrogen) by certain bacterial species like Rhizobium, Azotobacter, Cyanobacteria, Azospirillum, etc. and by other natural phenomena.

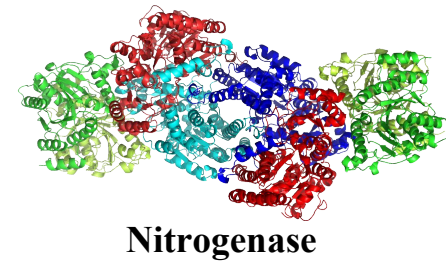
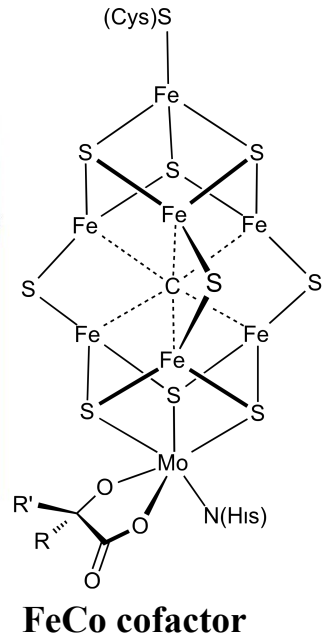
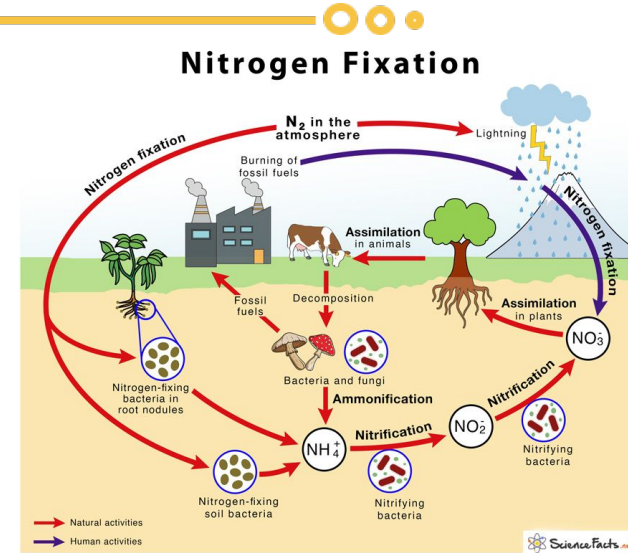
Plants obtain nitrogen from the soil and utilize it in the process of protein synthesis. Unlike carbon dioxide and oxygen, atmospheric nitrogen cannot be obtained through the stomata of leaves. Because the nitrogen gas present in the atmosphere can not be directly used by plants. There are certain bacteria and some natural phenomenon that help in Nitrogen fixation.

Molecular nitrogen consists of two nitrogen atoms joined by a triple covalent bond, making it highly inert and nonreactive. To break this bond and add three hydrogen atoms, the following reaction occurs:

$$\text{N}_2 + 8\text{H}^+ + 8\text{e}^- + 16 \text{ATP} \rightarrow 2\text{NH}_3 + \text{H}_2 + 16 \text{ADP} + 16\text{P}_i$$

The enzyme nitrogenase consists of two soluble proteins: component I and component II. Component I, known as MoFe protein, consists of 2 Mo atoms, 28 to 34 Fe atoms, and 26 to 28 sulfides, together known as an iron-molybdenum cofactor (FeMo cofactor). Component I is composed of two copies of subunits α and β . Component II, known as nitrogenase reductase, is composed of two copies of a single subunit. This protein contains 4 Fe atoms and four acid-labile sulfides (4Fe-4S).

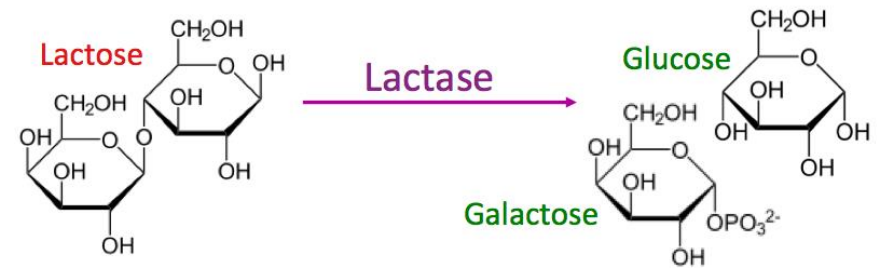
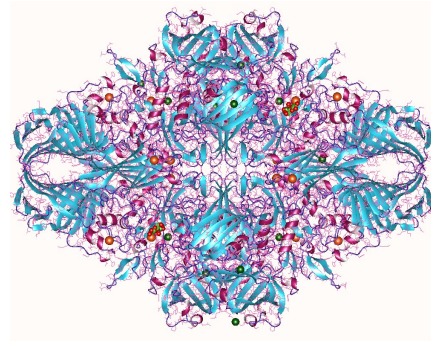
Component I, where substrate binding and reduction occur, binds to ATP and ferredoxin or flavodoxin proteins (Fdx or Fld). The energy for transferring electrons by Fdx or Fld comes from the hydrolysis of ATP. It is a reduction reaction, where electrons are added to N_2 to form NH_3 . Thus, component II's role is to actively transfer electrons, one at a time, to the component I, coupled to the hydrolysis of ATP.



Some important biocatalysts and their activities

- **Lactase- the curdler.**

- Converts milk into curd.
- Present throughout prokaryotes to eukaryotes.
- Used as supplements for lactose intolerant people.

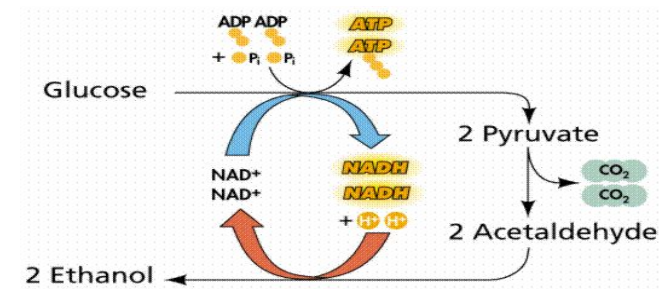


- **Alcohol dehydrogenase- the winemaker.**

- Catalyzes the fermentation process.
- Helps conversion of carbohydrates into alcohols by bacteria.
- In humans, it helps the metabolism of alcohol in liver.



<http://doi.org/10.2210/pdb1A4U/pdb>

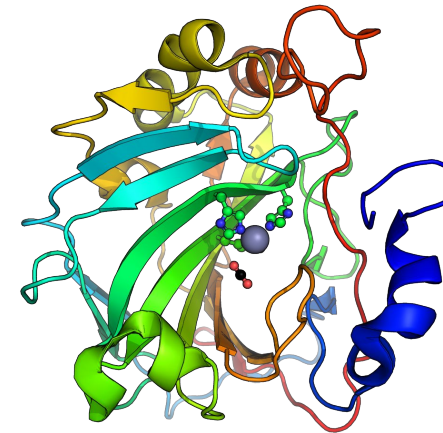


Some important biocatalysts and their activities

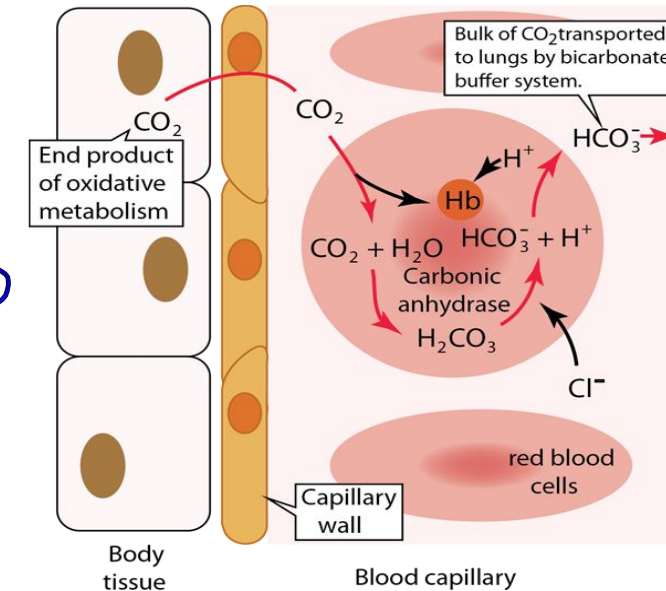


Carbonic anhydrase- Carbon dioxide excretor.

1. Present in RBC membranes.
2. Only **15 %** of carbon dioxide can be removed from the cells of our body **through blood plasma**.
3. Maintains acid-base balance.
4. Helps in translocation of around **85%** of carbon dioxide **from cells to lungs**
5. Catalyzes the hydration of carbon dioxide to form carbonic acid and rapidly dissociate into water.



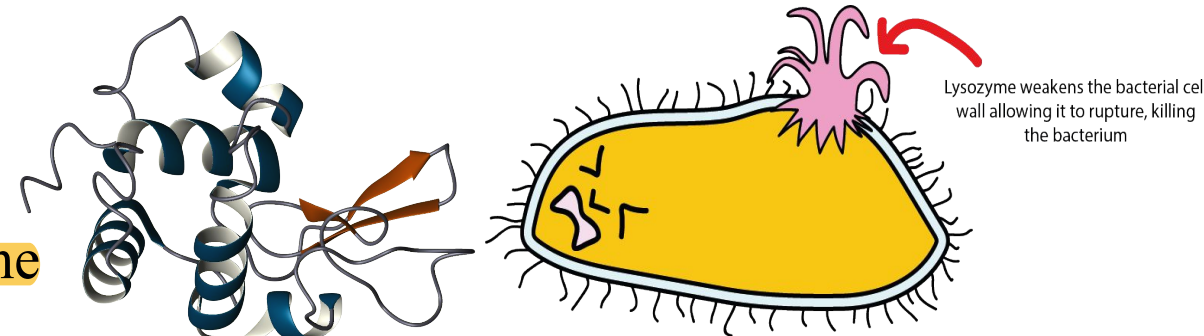
<http://doi.org/10.2210/pdb6LUX/pdb>



<http://hyperphysics.phy-astr.gsu.edu/hbase/Organic/carbanh.html>

Lysozyme- Body's own bacterial terminator.

1. Present in body fluids like **saliva, tears** etc.
2. Can catalyze the **breakage of bonds present in the cell wall of bacteria**. It is an **antimicrobial enzyme**



<https://bmr.bio/featuredSys/Lysozyme/lysozyme1.shtml>

<https://www.pngwing.com/en/free-png-xwnuy/download>

MOST (BUT NOT ALL) ENZYMES ARE PROTEINS, BUT ALL PROTEINS ARE NOT ENZYMES!!

Growth hormones, insulin, collagen, keratin – they are proteins but not enzymes

- Carl Neuberg
(1908)

RIBOZYMES- RNAs turned enzymes

- RNAs are information containing molecules just like DNA.
- However, they have a secondary function which was discovered much later after their discovery.
- **OH** groups present in their structure aid in their catalysis.

